ENGINEERING WORKSHOP

Project Title:

Robotic Arm (6 DOF)

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Table of Contents

1. ABSTRACT*:*

This project presents the design and development of a 6 Degrees of Freedom (DOF) robotic arm controlled by a miniature robotic arm with integrated potentiometers for position feedback. The system is designed to simulate and replicate the movements of the main robotic arm based on the inputs received from the mini arm, enabling precise control and manipulation. The main robotic arm is modeled in SolidWorks, allowing for detailed mechanical design and motion analysis. Potentiometers attached to the mini arm measure joint angles and provide real-time feedback to a microcontroller, which processes the data to control the movement of the main arm. The project integrates hardware components, including motors, sensors, and microcontrollers, with software algorithms for calculations and motion control. The result is a functional, interactive robotic system capable of replicating the movements of the mini arm, offering valuable insights into the fields of robotics, control systems,

and automation. This design can serve as a foundation for more advanced applications in robotic manipulation and automation tasks**.**

2. INTRODUCTION:

Robotic arms have become a cornerstone of automation and precision tasks across various industries, including manufacturing, healthcare, and research. The ability to control the position and movement of these arms with high accuracy and flexibility is crucial for achieving complex operations. This project focuses on the design and development of a 6 Degrees of Freedom (DOF) robotic arm that is controlled using a miniature robotic arm equipped with potentiometers for precise joint position feedback. The mini arm acts as a control interface, where the movement of its joints is sensed and translated into corresponding actions in the main robotic arm.

The main robotic arm, designed and modeled using SolidWorks, consists of six independently controlled joints, allowing for a wide range of motion and flexibility. Each joint is connected to a motor, with feedback from potentiometers attached to the mini arm's joints providing real-time positional data. This data is processed by a microcontroller to calculate the necessary movements for the main arm, ensuring accurate replication of the mini arm's movements.

This project combines mechanical engineering, control systems, and electronics to create an interactive robotic arm. The control system, using a microcontroller and potentiometer feedback, maps the motion of a mini arm to the main arm, enabling precise manipulation. The goal is to demonstrate a system that mimics the movement of the smaller arm with high accuracy, offering insights for future robotics research and applications. This design can be adapted for various robotic tasks, enhancing the control and manipulation of multi-jointed robotic systems.

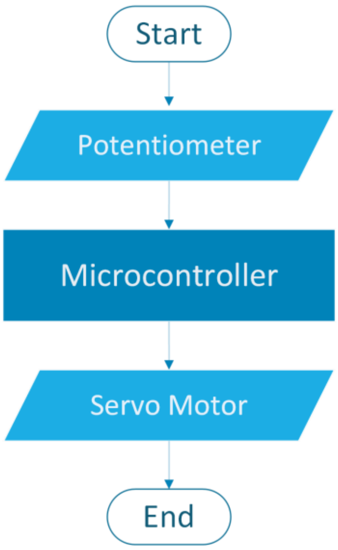
3. SOFTWARE TOOLS

The following software tools are commonly used in the design, and control of robotic systems like the one described in this

- **SolidWorks:** Used for 3D modeling and design of the robotic arm, including the main arm, joints, and mini arm. SolidWorks also enables motion simulations and interference checks. CAD design, motion simulation, assembly design, kinematic analysis.

- **Arduino IDE:** The primary environment for programming the microcontroller (such as Arduino or ESP32) that controls the robotic arm. C/C++ programming, serial communication, support for sensors and actuators, integration with external libraries for controlling motors and reading potentiometer values.
- **Proteus:** Proteus supports simulation of microcontrollers, so you can write the control code (e.g., Arduino code) in the Arduino IDE, then upload it to the Proteus simulation for testing. PCB design, schematic capture, and layout. The PCB design for this project was created with the assistance of **Engr. Zaryab Quazi**, who provided valuable expertise in the design and layout of the circuit board.

4. FLOW CHART



5. CODE

```
#include <Servo.h>
```

```
 Servo servo1;  const int
servo1PotPin =A0;  const
int servo1Pin = 7;  int
servo1Value;  Servo servo2;
const int servo2PotPin =A1;
const int servo2Pin = 6;
int servo2Value;  Servo
servo3; const int
servo3PotPin =A2; const int
servo3Pin = 5; int
servo3Value;  Servo servo4;
const int servo4PotPin =A3;
const int servo4Pin = 3;
int servo4Value;  Servo
servo5; const int
servo5PotPin =A4; const int
servo5Pin = 4; int
servo5Value;
Servo servo6;
const int servo6buttonPin =13;
const int servo6Pin = 2; int
pushButton;  void setup() {
Serial.begin(9600);
servo1.attach(servo1Pin);
servo2.attach(servo2Pin);
servo3.attach(servo3Pin);
servo4.attach(servo4Pin);
servo5.attach(servo5Pin);
servo6.attach(servo6Pin);
}  void loop() {   servo1Value =
analogRead(servo1PotPin);    servo1Value =
map(servo1Value, 1023, 0, 0, 180);
servo1.write(servo1Value);
Serial.print(servo1Value);
   servo2Value = analogRead(servo2PotPin);
servo2Value = map(servo2Value, 0, 1023, 0, 180);
servo2.write(servo2Value);
  Serial.print("   ");Serial.print(servo2Value);
     servo3Value = analogRead(servo3PotPin);
servo3Value = map(servo3Value, 0, 1023, 0, 180);
servo3.write(servo3Value);
  Serial.print("   ");Serial.print(servo3Value);
   servo4Value = analogRead(servo4PotPin);
servo4Value = map(servo4Value, 0, 1023, 0, 180);
servo4.write(servo4Value);
  Serial.print("   ");Serial.print(servo4Value);
   servo5Value = analogRead(servo5PotPin);
servo5Value = map(servo5Value, 0, 1023, 0, 180);
servo5.write(servo5Value);
```

```
  Serial.print("   ");Serial.print(servo5Value);

pushButton=digitalRead(servo6buttonPin);
if(pushButton==LOW)
  {
    servo6.write(0);
    Serial.print("   ");Serial.print("Close");
  }
else
{
    servo6.write(45);
    Serial.print("   ");Serial.println("Open");
  }
  delay(5);
}
```

6. WORKING

The robotic arm operates through the input of six control elements: five potentiometers and a push button. These control elements are used to monitor and adjust the position of the arm's joints. Each potentiometer is connected to a specific joint on the mini robotic arm and acts as a position sensor, determining the angle of the joint. When the mini arm moves, the corresponding potentiometer detects the change in joint position by varying its resistance. This change in resistance is converted into an analog voltage, which is then sent to the microcontroller for processing.

The microcontroller reads the analog values from the potentiometers via the Analog Read Pins. Since the potentiometer's output ranges from 0 to 1023, corresponding to the full range of motion (0 to 180 degrees) of the servo motors, the microcontroller maps these values to a servo-friendly range using the map() function. This function adjusts the range of values from (0-1023) into a 0–180-degree range, which is the operational range for standard servos.

Once the potentiometer values are mapped to a degree range, the microcontroller sends the corresponding signals to the servo motors, which adjust the position of the joints in the main robotic arm.

The robotic arm includes a push button that controls the sixth servo, which might represent a gripper or similar mechanism. This push button is connected to a digital input pin of the microcontroller. The button has two possible states: pressed (LOW) or released (HIGH). When the button is pressed (LOW state), the sixth servo moves to the "closed" position, which could mean the gripper closes. When the button is released (HIGH state), the sixth servo moves to the "open" position, causing the gripper to open. This allows for manual control of a specific function, such as gripping or releasing objects.

The serial monitor displays the potentiometer values in real-time, providing feedback on each joint's angle as the potentiometers are adjusted. This helps the user track the arm's movements and troubleshoot if necessary. A 5-millisecond delay in the loop() function ensures smooth and controlled motion, preventing the servos from moving too quickly and causing instability.

In summary, the system reads potentiometer inputs to control the servo motors of the main robotic arm, with the push button controlling a sixth servo (e.g., a gripper). The serial monitor provides real-time feedback, offering an intuitive way to control the robotic arm based on the mini arm's movements.

7. RESULT AND DISCUSSIONS

The 6-DOF robotic arm operated as expected, with the potentiometers providing accurate control over the arm's movements. Each joint's position was directly linked to its corresponding potentiometer, with smooth and responsive servo motion. The push button effectively controlled the 6th servo, allowing for simple gripper functionality, toggling between "open" and "closed" positions.

The 5-millisecond delay in the loop() function ensured smooth motion and prevented jittery movements by slowing down the servo response. The real-time feedback on the serial monitor helped track the arm’s performance, providing useful data for troubleshooting.

Overall, the system performed well with precise and synchronized movement between the mini and main robotic arms. Potential improvements include fine-tuning the servo precision, adding advanced control algorithms, and incorporating additional sensors for enhanced functionality.

8. CONCLUSION

The 6-DOF robotic arm, controlled by potentiometers and a push button, successfully demonstrated the ability to replicate the motion of a mini robotic arm. The system operated smoothly, with the potentiometers providing precise control over each joint, and the push button enabling simple control of the 6th servo (e.g., for gripper function). The use of a delay ensured stable movement, and the serial monitor provided real-time feedback for monitoring and debugging.

This project highlights the potential for using basic sensors and simple control systems to operate complex robotic systems. Future improvements could involve refining the control algorithms and adding more sensors for enhanced precision and functionality. Overall, the project serves as a solid foundation for further exploration and development in robotic control systems.

9. PROJECT SUMMARY

This project involves the design and implementation of a 6-DOF robotic arm controlled by a mini arm equipped with potentiometers and a push button. The potentiometers detect the angle of each joint on the mini arm, and their corresponding values are sent to a microcontroller, which processes these inputs to control the movement of the main robotic arm. The push button controls a 6th servo, such as a gripper, allowing for simple open/close functionality. The system operates smoothly, with real-time feedback displayed on the serial monitor for monitoring the arm's movements. A delay is included in the control loop to ensure stable motion and avoid erratic servo movements. This project demonstrates the use of basic sensors and control systems to create a functional and responsive robotic arm, offering a foundation for further development in robotic automation.

10. REFERENCES

Arduino. (n.d.). *Arduino Documentation*. Retrieved December 26, 2024, from https://www.arduino.cc/